B. Asterisk-Minor Template

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: implementation, strings | https://codeforces.com/problemset/problem/1796/B

You are given two strings *a* and *b*, consisting of lowercase Latin letters.

A template *t* is string, consisting of lowercase Latin letters and asterisks (character '*'). A template is called *asterisk-minor* if the number of asterisks in it is less than or equal to the number of letters in it.

A string *s* is said to be matching a template *t* if you can replace each asterisk in *t* with a string of lowercase Latin letters (possibly, an empty string) so that it becomes equal to *s*.

Find an *asterisk-minor* template such that both *a* and *b* match it, or report that such a template doesn't exist. If there are multiple answers, print any of them.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of testcases.

The first line of each testcase contains a string a ($1 \le |a| \le 50$, where $|a|$ is the length of a), consisting of lowercase Latin letters.

The second line contains a string b ($1 \le |b| \le 50$), consisting of lowercase Latin letters.

Output

For each testcase, output "NO", if there doesn't exist an *asterisk-minor* template that both *a* and *b* match. Otherwise, print "YES" in the first line and the template in the second line. If there are multiple answers, print any of them.

A template should consist only of lowercase Latin letters and asterisks (character '*'). The number of asterisks should be less than or equal to the number of letters.

Example

Input
6 aaab zzzb codeforces atcoder codeforces tokitlx aaaa aaaaaa abcd abcd c f

Output
YES *b YES *co* NO YES a*a*a*a YES abcd NO

Note

In the first testcase, for a template "*b", you can replace the only asterisk with "aaa" to get "aaab" (which is equal to *a*) or with "zzz" to get "zzzb" (which is equal to *b*).

In the third testcase, a template "*o*" is not *asterisk-minor*, as it contains more asterisks than letters. There are no *asterisk-minor* templates that both *a* and *b* match.

In the fourth testcase, for a template "a*a*a*a", you can replace all asterisks with empty strings to get "aaaa" (which is equal to *a*) or two of them with "a" and two of them with an empty string to get "aaaaaa" (which is equal to *b*).

In the fifth testcase, there are no asterisks in a template "abcd", so only "abcd" can match it (which is coincidentally both *a* and *b*).